

Rebuttal for CoRL Submission #1453

We thank the AC and reviewers for their thoughtful feedback. Below, we address the concerns raised in the reviews.

@AC @pWgy I. Geometry. We use a right-handed robot-local frame with x along head-yaw forward, y left, and z up. Each object contributes a predicted 3-D center (x, y, z) with visibility confidence. Both heads use only the current 32-D depth latent, with head states provided separately through actor proprioception. Visibility labels use the full camera transform and valid field-of-view/depth-range constraints: a single-point object is visible if its reference point is valid, and a multi-point object if any predefined structural point is valid, while geometry always predicts only the object center. During training, GT and predicted geometry are mixed as $\tilde{\mathbf{y}}_t^i = (1 - \beta_k)\mathbf{y}_t^i + \beta_k\hat{\mathbf{y}}_t^i$, where β_k increases from 0 to 1. The actor receives $[c_t^i\tilde{\mathbf{y}}_t^i, c_t^i]$, where $c_t^i = \text{clip}(\text{stopgrad}(\tilde{p}_t^i), 0.05, 1)$. At deployment, all auxiliary features are predicted from depth.

@AC @pWgy @f48j II. Ablations. Measured perception quality. Within typical task ranges, mean bearing/range MAEs are $2.5^\circ/0.022$ m for dribbling and $6.06^\circ/0.184$ m and $8.16^\circ/0.266$ m for the shooting ball and goal. At longer ranges, dribbling error rises about $20\times$, mainly from range underestimation, and shooting-ball error about $1.5\times$. In shooting, ball error is $4.57\times$ higher at the image boundary than center and rises $1.28\times$ at 45° – 60° head yaw; goal error changes by under 2%. Dribbling contact raises bearing error $1.5\times$ briefly, recovering within 0.3 s. Visibility errors concentrate in extreme conditions: beyond 2.3 m, dribbling misclassifies 84% of invisible samples as visible; under extreme head poses, 78% of shooting predictions labeled visible are incorrect. Overall, the auxiliary heads are reliable in normal task ranges but degrade under geometric and visibility stress. **Auxiliary head.** Each ablation changes one Full switch: *w/o anneal* keeps $\alpha=0$ (always predicted aux, no GT→pred schedule); *w/o geometry* zeros Actor aux input while keeping aux losses; *w/o supervision* drops goal/ball/visibility aux losses but still feeds predicted aux; *w/o vis gate* skips visibility×geometry masking. All components help SR; Full is best. **Active vision.** Its internal components are not separable for component-wise ablation because they all act through the same two head-joint control dimensions. Thus, active vision can only be ablated as a whole by setting these two action scales to zero.

Methods	Penalty Kick			Free Kick			S-Slalom		
	SR $\uparrow$	$E_a\downarrow$	$r_c\uparrow$	SR $\uparrow$	$E_a\downarrow$	$r_c\uparrow$	SR $\uparrow$	$E_v\downarrow$	$r_b\uparrow$
DAVIS	0.84	23.29	99.75	0.80	30.18	99.27	0.70	0.79	0.74
w/o anneal	0.32	54.79	84.87	0.22	76.20	87.92	0.31	0.83	0.66
w/o geometry	0.20	53.53	87.33	0.15	69.47	78.13	0.34	0.91	0.54
w/o supervision	0.13	81.87	93.33	0.14	93.66	91.67	0.30	0.95	0.48
w/o vis gate	0.34	51.97	91.33	0.29	68.44	92.08	0.27	1.06	0.64
heuristic head controller	0.88	26.66	93.67	0.76	37.96	88.13	0.0	0.99	0.13

@AC @f48j @rFts @f6aS III. Comparisons. Comparative setting. DAVIS maps onboard depth frames directly to control. Main-paper references [25] and [28] instead use additional sensing/localization: [25] uses mocap robot/ball states, restricting deployment to a mocap-equipped field, while

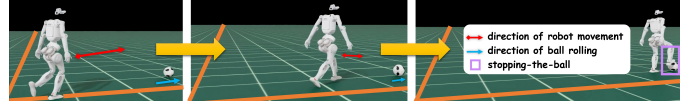


Fig. 1: Simulation frames of stopping-the-ball skill, instantiated during rebuttal by modifying only the task definition. [28] combines RGB-D detection with LiDAR global localization to recover an out-of-view ball. No same-robot modular baseline is available, so we use the architecturally closest published system. DAVIS achieves comparable success rates to Dribble Master [1] on both core tasks: target reaching (0.88 vs. 0.87) and obstacle avoidance (0.96 vs. 0.93). **Head controller.** We also compared end-to-end learned head control with a heuristic that gazes at the predicted ball, as shown in the table. For static-object tasks such as penalty kicks, the heuristic head controller performs comparably to active vision. However, in highly dynamic dribbling scenarios, active vision shows a clear advantage. **Depth-only setting.** DAVIS uses only head-mounted depth and proprioception for end-to-end humanoid soccer control. Its tasks rely mainly on local ball/goal geometry directly supervised in simulation, and our results show depth suffices under the evaluated conditions.

@AC @f48j @f6aS IV. Robot trials. Trials and protocol. Shooting achieved 26/38, 28/49, and 33/60 successes for Easy/Medium/Hard; dribbling achieved 22/28, 19/28, and 17/28. Shooting used progressively harder initializations within one penalty-kick protocol, with 38/49/60 trials to characterize the harder conditions; each dribbling setting had 28 trials. App. B.4 details task settings and success criteria. **Failures and safety.** The dominant failures were loss of task-relevant objects from the depth view and overly forceful contacts that drove the ball out of bounds. After full ball loss, the visibility-aware active-vision policy (App. F.3) decelerates and reorients the head/body to reacquire it. Otherwise, control returns to default walk–run policy, which stably tracks only joystick velocity commands. Training regularizes posture, contact, and joints; safeguards include an emergency stop, fall trigger, and tether/staff protection on grass. A safety-interrupted rollout is repeated from the same initial condition.

@pWgy @rFts @f6aS V. Deployment commands and task instantiations. Joystick. Shooting actor receives no operator command; in dribbling, the joystick supplies only intended direction. **Single policy.** Shooting and dribbling use separate task-specific checkpoints. **Extensibility.** Appendix F, the final supplementary-video simulation sequence, and Fig. 1 show a close-range powerful shot, obstacle-aware dribbling, ball-loss recovery, and **stopping-the-ball**. Together, they show DAVIS efficiently instantiates new soccer skills; our framework focuses on single-skill instantiation.

REFERENCES

- [1] Z. Wang, J. Zhou, and Q. Wu. Dribble master: Learning agile humanoid dribbling through legged locomotion, 2025.